

A NEW TREATMENT OF SINGULAR POTENTIALS

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For a broad class of strongly singular potentials, the effective (finite-dimensional) forms of the hamiltonian H are constructed by its "smooth" algebraic truncation in a large model space. The method is based on an asymptotic factorization of H into a product of matrices. Its efficiency (acceleration of convergence of the energies) is illustrated on the singularly anharmonic potential $V(r) = r^2 + hr^{-4}$.

In the various applications of quantum mechanics we often encounter phenomenological potentials which are strongly singular at the origin. For example, the Lennard-Jones forces between the molecules, nucleon-nucleon potentials with a core, singular models of fields in zero dimensions, etc. [1]. In the radial Schrödinger equation

$$[-d^2/dr^2 + l(l+1)/r^2 + V(r)] \psi(r) = E\psi(r),$$

$$l = 0, 1, \dots, \quad (1)$$

they may be approximated by their truncated power-law expansion

$$V(r) = V_{pq}(r) = \sum_{k=-p}^q g_k r^{2k}, \quad g_{-p} > 0, \quad p \geq 2, \quad (2)$$

and/or by its various $r \rightarrow r^{\text{const}}$ modifications [2].

In a purely mathematical context, a lot of difficulties related to the $r \rightarrow 0$ singularity of (2) has been described in the literature [1]. Our present intention is to demonstrate that at least some of them are not so serious in the bound-state problem (1), (2) with $q \geq 1$ and $g_q > 0$:

(i) We shall show that one of the most common eigenvalue methods, namely, an approximate solution of the Schrödinger equation in the truncated harmonic-oscillator basis, becomes easily applicable to (1), (2) after a slight modification. In this setting, the singular-

ity of (2) does not violate the applicability of some semi-numerical (e.g., matrix continued fractional [3]) methods designed originally for the regular $p = 0$ anharmonic oscillators only.

(ii) An introduction of a perturbative expansion of the exact, finite-dimensional effective hamiltonian [4] is one of the most current improvements of (i). Here, in essence, we shall derive a closed formula representing its asymptotic summ to all orders.

In more detail, let us start from the threshold behaviour

$$\psi(r) \sim \exp[-g_{-p}^{1/2} r^{1-p}/(p-1)], \quad r \sim 0,$$

of the physical solutions [2]. Obviously, the functions $r^{-p}\psi(r)$ remain regular at the origin – they may be expanded into the complete set $|n\rangle$ of the ordinary harmonic-oscillator eigenstates [3],

$$H_0 |n\rangle = \epsilon_n |n\rangle, \quad \epsilon_n = 4n + 2l + 3,$$

$$H_0 = -d^2/dr^2 + l(l+1)/r^2 + r^2,$$

$$|r^{-p}\psi\rangle = \sum_{n=0}^{\infty} |n\rangle z_n.$$

As a consequence, we may rewrite our Schrödinger equation (1), (2) in the algebraic form

$$\sum_{n=0}^{\infty} \mathcal{H}_{mn} z_n = 0, \quad m = 0, 1, \dots, \quad (3)$$

where

$$\mathcal{K} = T^s K_0 T^s + \sum_{k=-p}^q g_k T^{p+k} - E T^p - T^{p+1}, \quad (4)$$

with

$$\begin{aligned} K_0 &= H_0, & p &= 2s, \\ &= \frac{1}{2}(H_0 T + T H_0) + 1, & p &= 2s+1, \\ s &= 0, 1, \dots, \end{aligned}$$

$$\begin{aligned} T_{ij} &= \langle i | r^2 | j \rangle \\ &= a_i = \epsilon_i / 2, \quad i = j = 0, 1, \dots \\ &= b_k = [(k+1)(k+1+3/2)]^{1/2}, \\ &k = i = j - 1 \text{ (or } k = j = i - 1) = 0, 1, \dots \end{aligned}$$

Eq. (3) is solvable numerically by a standard numerical method, namely, by the diagonalization of the $(M \times M)$ -dimensional submatrices of $\mathcal{K}$ in the limit $M \rightarrow \infty$. A sample of the results obtained for the simplest nontrivial potential $V_{21}(r) = g_{-2}r^{-4} + g_{-1}r^{-2} + g_0 + g_1r^2$ is given in the first column of table 1 here.

Table 1
Ground state energies in $V(r) = r^2 + hr^{-4}$ and the acceleration of their convergence.

h	M	Ordinary truncation	Smooth truncation (9)
0.02	10	3.466	3.358
	30	3.327	3.290
	50	3.300	3.281
	100	3.285	3.281
	300	3.283	3.283
	500	3.282	3.282
0.04	10	3.509	3.419
	30	3.403	3.383
	50	3.390	3.384
	100	3.386	3.386
	300	3.386	3.385
	500	3.385	3.385
0.4	10	4.03906	4.03841
	30	4.03579	4.03366
	50	4.03312	4.03204
	100	4.03202	4.03195
	300	4.03198	4.03197
	500	4.03197	4.03197

As expected, the rate of convergence of the ordinary truncation method is rather slow — this follows from the marked difference between $\psi(r)$ and $\langle r | n \rangle$ at $r \sim 0$, as well as from the observation that $|z_n/z_{n+1}| \approx 1$ for $n \gg 1$ and $p = 0$ at least [5]. Hence, the exact effective $(M \times M)$ -dimensional form of (3) [4]

$$\sum_{n=0}^{M-1} \mathcal{K}_{mn}^{(\text{eff})} z_n = 0, \quad m = 0, 1, \dots, M-1, \quad (5)$$

should be considered, with the definition

$$\mathcal{K}^{(\text{eff})} = P \mathcal{K} P - P \mathcal{K} Q (Q \mathcal{K} Q)^{-1} Q \mathcal{K} P, \quad (6)$$

$$P = 1 - Q = \sum_{n=0}^{M-1} |n\rangle \langle n|,$$

employed at least in an approximate way. In the present context, it has the following two merits:

(m₁) In the light of (4), $\mathcal{K}$ is a band, $2t+1$ — a diagonal in general — we get

$$\mathcal{K}_{ij}^{(\text{eff})} = \mathcal{K}_{ij}, \quad i, j = 0, 1, \dots, M-1,$$

$$(i, j) \neq (M-m, M-n), \quad m, n = 1, 2, \dots, t, \quad (7)$$

directly from the definition (6). Thus, the exact equation (5) does not differ “too much” from the truncated forms of (3).

(m₂) The matrix elements $T_{M+i, M+j} = M \times \binom{2}{1+i-j} + O(1)$ increase quickly with growing M . Hence, we may assume that the finite, P -projected model space is large, $M \gg 1$, and consider the truncated asymptotic expansions of $Q \mathcal{K} Q$ only.

Let us restrict our attention here to the leading-order asymptotic contributions only, $Q \mathcal{K} Q \approx Q T^t Q$, $t = p + q - \delta_{q1}$ [for $q = 1$, the scaling $r \rightarrow \text{const} \times r$ gives $g_1 \rightarrow 1$ in (4)]. Then, introducing the notation $Q_{i-1} T = Q_{i-1} T Q_i = H_i$,

$$Q_i = \sum_{n=M-i}^{\infty} |n\rangle \langle n|,$$

$i = 1, 2, \dots, M$, the factorizability of $Q T^t Q$ gives, in a highly formal manner, the cancellations

$$\begin{aligned} P \mathcal{K} Q (Q \mathcal{K} Q)^{-1} Q \mathcal{K} P &\approx P H_t^T H_{t-1}^T \dots H_1 \\ &\times (H_1 H_2 \dots H_m K H_m^T H_{m-1}^T \dots H_1^T)^{-1} H_1 H_2 \dots H_t P \\ &\approx P H_t^T H_{t-1}^T \dots H_{m+1}^T K^{-1} H_{m+1} H_{m+2} \dots H_t P, \quad (8) \end{aligned}$$

where $t = 2m$ and $K = I$, or $t = 2m + 1$ and $K = T$. With the particular matrix elements as given by (4), the leading-order formula (8) may be further reduced. Indeed, after simple binomial multiplications with

$$H_1 H_2 \dots H_m \approx H_1^m$$

$$\approx M^m \begin{pmatrix} \binom{2m}{0} \binom{2m}{1} \dots \binom{2m}{2m} 0 0 \dots \\ 0 \quad \binom{2m}{0} \dots \binom{2m}{2m-1} \binom{2m}{2m} 0 \dots \\ \vdots \end{pmatrix},$$

etc., we get finally the compact $M \gg 1$ prescription

$$\mathcal{H}_{M-i M-j}^{(\text{eff})} = \mathcal{H}_{M-i M-j} \times \binom{2t}{t+i-j}^{-1} \sum_{k=1}^t \binom{t}{i-k} \binom{t}{j-k},$$

$$i, j = 1, 2, \dots, t, \quad (9)$$

complementing (7) and completing our asymptotic

algebraic definition of the exactly solvable eigenvalue condition (5), $\det \mathcal{H}^{(\text{eff})} = 0$.

In the same way as above, the $M \gg 1$ numerical algorithm (5), (7), (9) has been tested (second column in table 1). The approximate energies converge, with increasing M and possible oscillations, to the exact variational values. Obviously, the "smooth cut-off" (9) causes an extremely efficient acceleration of convergence. A systematic inclusion of the higher-order corrections is under current investigation at present.

References

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